

Ex. No. : 10

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A PYTHON PROGRAM TO IMPLEMENT DIMENSIONALITY REDUCTION – PCA

Aim:

To implement **Principal Component Analysis (PCA)** for dimensionality reduction on the Iris dataset.

Requirements:

Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler, LabelEncoder

from google.colab import files
uploaded = files.upload()

df = pd.read_csv('Iris.csv')

X = df[['SepalLengthCm','SepalWidthCm','PetalLengthCm','PetalWidthCm']]
y = df['Species']

le = LabelEncoder()
y = le.fit_transform(y)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print("Explained Variance Ratio:", pca.explained_variance_ratio_)
```

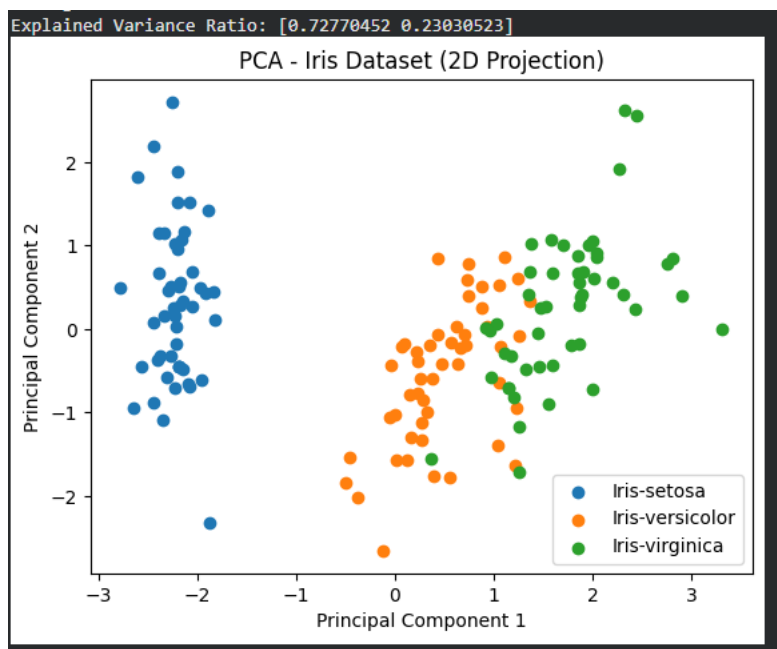
```
plt.figure()

for i in np.unique(y):
    plt.scatter(
        X_pca[y == i, 0],
        X_pca[y == i, 1],
        label=le.inverse_transform([i])[0]
    )

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA - Iris Dataset (2D Projection)")
plt.legend()

plt.show()
```

OUTPUT:



Result:

Thus, PCA successfully reduced the dataset dimensions while retaining important variance, enabling better visualization and analysis.